

STEGANOGRAPHY ANALYSIS AND HIDDEN DATA DETECTION USING STEGEXPOSE

EX. NO: 8

Date :

1. AIM

To analyze digital image evidence using the StegExpose command-line utility, evaluate LSB-based statistical properties and generate a structured CSV report for forensic documentation.file.

2. SOFTWARE / TOOLS REQUIRED

Item	Details
Operating System	Windows
Runtime Environment	Java Runtime Environment (JRE)
Java Version	26.0.2.1
Forensic Tool	StegExpose Java .jar executable
StegExpose Version	0.1
Evidence Type	Digital image evidence
Evidence File	spider-man.png
Evidence Format	PNG
Evidence Size	23,264 bytes
Analysis Mode	default
Detection Threshold	0.2
Output Report	report.csv
Report Size	248 bytes
Report Viewer	Microsoft Excel / Notepad

Utilities / Components Used

- **Java Runtime Environment (JRE)** – Used to execute the StegExpose Java application.
- **StegExpose.jar** – Used to perform statistical steganalysis of the image evidence.
- **Evidence directory** – Contains the image subjected to analysis.
- **report.csv** – Contains the structured StegExpose analysis results.
- **Microsoft Excel / Notepad** – Used to review the generated CSV report.

3. THEORY

StegExpose is a specialized steganalysis tool designed to detect the presence of hidden data within digital images. Unlike traditional steganography tools used to embed information, StegExpose serves an investigative purpose by analyzing statistical properties of digital images for anomalies that may indicate data hiding.

The tool analyzes image characteristics associated with steganographic techniques, particularly Least Significant Bit (LSB) based data hiding. StegExpose produces quantitative statistical values that can be evaluated against a configured detection threshold.

4. PROCEDURE

Step 1: Working Directory and Java Verification

The Windows PowerShell environment was used to access the Experiment 08 working directory.

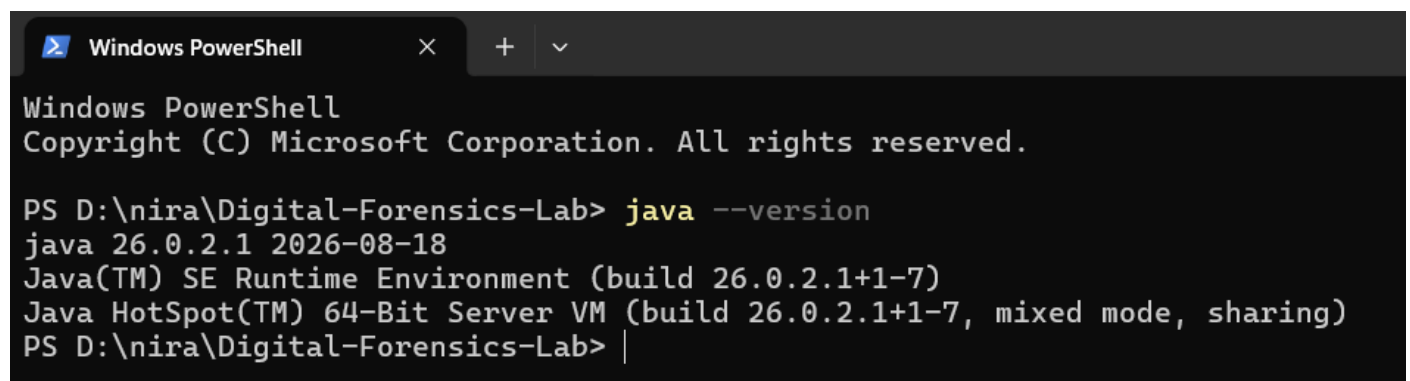
The experiment directory was:

D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose

The Java Runtime Environment was verified using: `java --version`

The installed Java version was: `java 26.0.2.1`

The working directory was also verified to contain the StegExpose application and experiment files.

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with a blue icon, a close button, and a dropdown menu. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', 'PS D:\nira\Digital-Forensics-Lab> java --version', 'java 26.0.2.1 2026-08-18', 'Java(TM) SE Runtime Environment (build 26.0.2.1+1-7)', 'Java HotSpot(TM) 64-Bit Server VM (build 26.0.2.1+1-7, mixed mode, sharing)', and 'PS D:\nira\Digital-Forensics-Lab> |'.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\nira\Digital-Forensics-Lab> java --version
java 26.0.2.1 2026-08-18
Java(TM) SE Runtime Environment (build 26.0.2.1+1-7)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.2.1+1-7, mixed mode, sharing)
PS D:\nira\Digital-Forensics-Lab> |
```

Figure 1: Verification of the Java runtime environment and StegExpose working directory.

Step 2: Evidence Directory Verification

The evidence directory was examined to verify the image file available for steganalysis.

The evidence directory contained: `spider-man.png`

The file size was: 23,264 bytes

The evidence was verified using the Windows command-line environment before executing StegExpose.

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose>dir ".\Evidence"
Volume in drive D is New Volume
Volume Serial Number is D05D-A148

Directory of D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose\Evid
ence
25-09-2026 11:14 <DIR> .
25-09-2026 11:18 <DIR> ..
25-09-2026 11:12 23,264 spider-man.png
1 File(s) 23,264 bytes
2 Dir(s) 79,835,459,584 bytes free
```

Figure 2: Verification of the evidence directory containing the spider-man.png image evidence.

Step 3: Execution of StegExpose Analysis

The StegExpose Java application was executed from the Experiment 08 working directory.

The command used was: `java -jar .\StegExpose.jar ".\Evidence" default 0.2 ".\report.csv"`

The command specifies:

- Evidence directory: `.\Evidence`
- Execution mode: `default`
- Statistical threshold: `0.2`
- Output report: `.\report.csv`

The StegExpose process completed successfully.

The PowerShell exit-code verification was performed using: `echo $LASTEXITCODE`

The returned exit code was: 0

indicating successful execution of the Java process.

```
Windows PowerShell x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> cd "D:\nira\Digital-Forensics-Lab\Ex-08-Steganogra
phy-Analysis-and-Hidden-Data-Detection-Using-StegExpose"
PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> java -jar .\StegExpose.jar ".\Evidence" default 0.
2 ".\report.csv"
PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> echo $LASTEXITCODE
0
PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> |
```

Figure 3: Execution of StegExpose on the evidence directory using default mode and a statistical threshold of 0.2.

Step 4: Generation and Verification of the CSV Report

After the StegExpose analysis completed, the generated report was verified using: `dir ".\report.csv"`

The report was successfully created as: `report.csv`

The file size was: 248 bytes

The contents were then displayed using: `Get-Content ".\report.csv"`

The generated CSV contained the actual StegExpose analysis result for `spider-man.png`.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> dir ".\report.csv"

Directory: D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose

Mode                LastWriteTime         Length Name
----                -
-a-----         25-09-2026  11:24 AM           248 report.csv

PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> Get-Content ".\report.csv"
File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
spider-man.png,false,412,0.0795903713184999,0.0,null,0.07967018891327492,0.053086853410591606
PS D:\nira\Digital-Forensics-Lab\Ex-08-Steganography-Analysis-and-Hidden-Data-Detection-Using-StegExpose> |

```

Figure 4: Verification of the generated report.csv file and its StegExpose analysis output.

Step 5: Review of the StegExpose Report

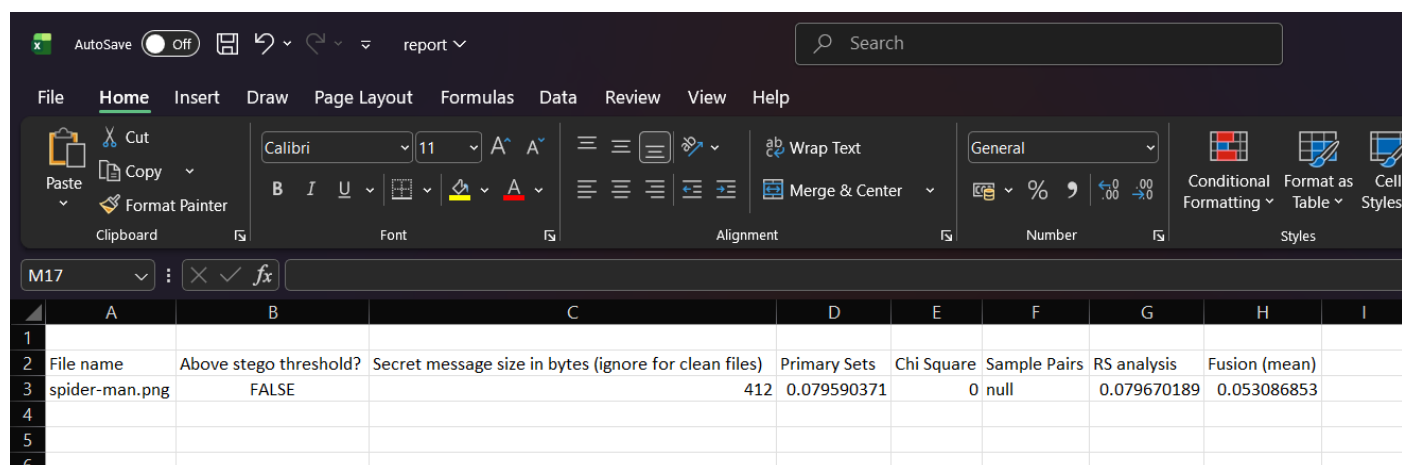
The generated report.csv file was opened using Microsoft Excel for detailed examination.

The Fusion (mean) value was: 0.053086853410591606

The configured threshold was: 0.2

The report indicated: Above stego threshold? = FALSE

Therefore, the analyzed image did not exceed the configured StegExpose threshold in this execution.



	A	B	C	D	E	F	G	H	I
1									
2	File name	Above stego threshold?	Secret message size in bytes (ignore for clean files)	Primary Sets	Chi Square	Sample Pairs	RS analysis	Fusion (mean)	
3	spider-man.png	FALSE	412	0.079590371	0	null	0.079670189	0.053086853	
4									
5									
6									

Figure 5: Review of the generated report.csv containing the StegExpose analysis results.

5. OBSERVATIONS / FORENSIC FINDINGS

1. Java 26.0.2.1 was successfully verified.
2. The evidence directory contained one image, spider-man.png.
3. StegExpose executed successfully with exit code 0.
4. The configured threshold was 0.2.
5. The Fusion (mean) score was 0.053086853410591606.
6. The report indicated Above stego threshold = FALSE.
7. The generated report.csv was successfully reviewed.
8. The Sample Pairs value was reported as null.

6. LIMITATION

StegExpose provides statistical indicators for steganalysis and does not by itself constitute conclusive proof of hidden data. The experiment analyzed only one image, and the Sample Pairs value was reported as null.

7. CONCLUSION

The experiment successfully demonstrated steganalysis using StegExpose. The image spider-man.png was analyzed using the default mode and a threshold of 0.2. A Fusion (mean) score of 0.053086853410591606 was obtained, and the report classified the image as not above the configured stego threshold.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

8.RESULT

The StegExpose steganalysis experiment was successfully completed. The evidence image was analyzed, the statistical results were generated, and report.csv was successfully created and reviewed. The Fusion (mean) score was **0.053086853410591606**, which was below the configured threshold of **0.2**.